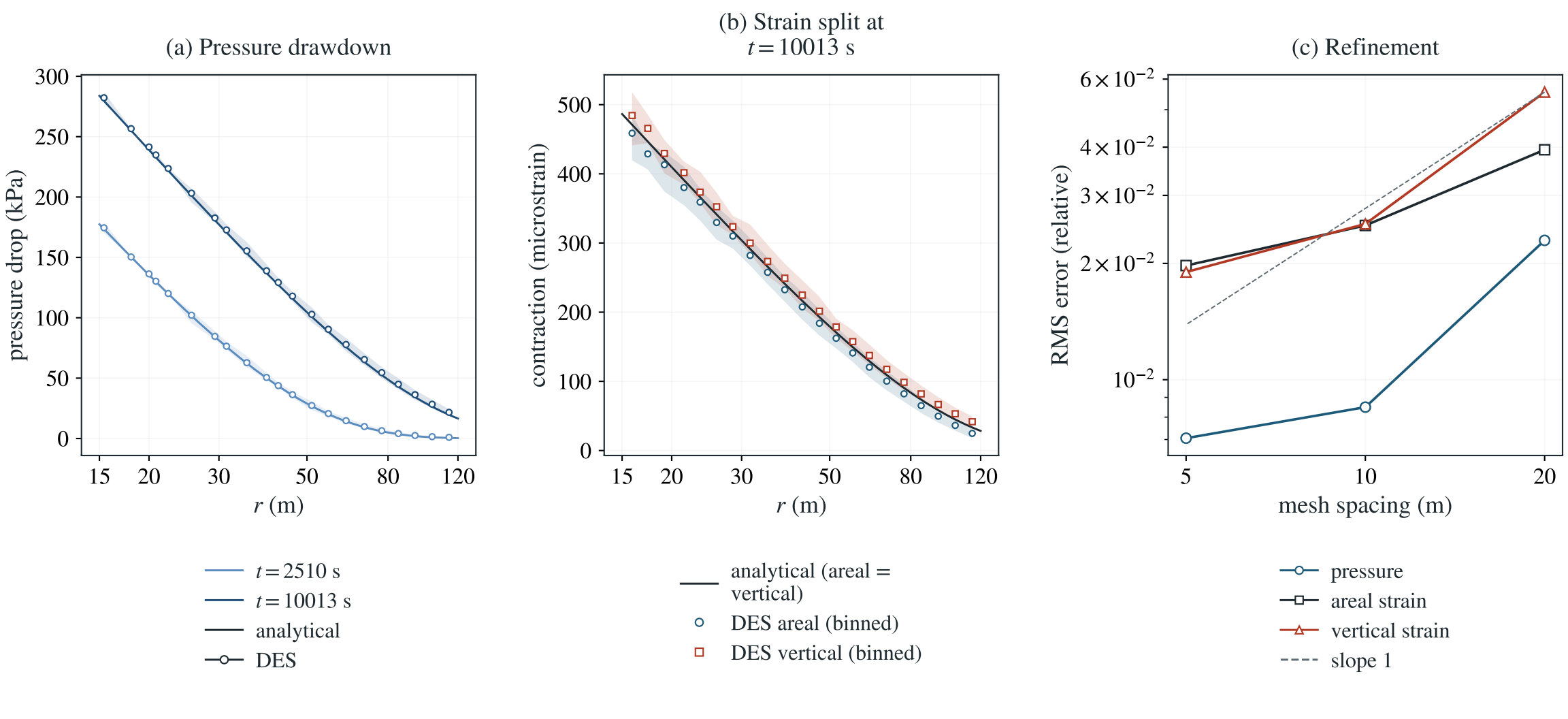

(a) Pressure drawdown
pressure drop (kPa)
300
250
200
150
100
50
0
15
20
30
50
80
120
r (m)
t = 2510 s
t = 10013 s
analytical
DES
(b) Strain split at
t = 10013 s
contraction (microstrain)
500
400
300
200
100
0
15
20
30
50
80
120
r (m)
analytical (areal = vertical)
DES areal (binned)
DES vertical (binned)
(c) Refinement
RMS error (relative)
6 × 10−2
4 × 10−2
3 × 10−2
2 × 10−2
10−2
5
10
20
mesh spacing (m)
pressure
areal strain
vertical strain
slope 1